

STA 35C – Homework 2

Submission due: Thu, October 8 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 2” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw2.pdf`). Include “STA 35C,” your name, and the last four digits of your student ID on the front page. For coding problems, include your R code, relevant output, and any requested plots/tables in the PDF. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full information about submission requirements and the late submission policy, see the syllabus.

Problem 1 (15 points in total).

- (a) (6 points) Suppose each of the 2 items has a $\frac{1}{3}$ chance of being defective, independently. Then

$$X \sim \text{Binomial}\left(2, \frac{1}{3}\right) \quad \text{and} \quad p_X(x) = \frac{2!}{x!(2-x)!} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{2-x}, \quad \forall x \in \{0, 1, 2\}.$$

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

(Hint: use the PMF or write X as the sum of two independent Bernoulli random variables.)

- (b) (9 points) Suppose that a manufacturing company ships products in boxes of **two** items each. Let X be the number of defective items in a box. Each box is produced by **Factory A** or **Factory B**, each with equal probability. Conditional on the factory, the two items are defective independently, with defect probability $\frac{1}{3}$ for Factory A and $\frac{1}{10}$ for Factory B.

Compute the probability that a randomly chosen box is from Factory A *and* has exactly one defective item. Then, given that a box has exactly one defective item, compute the probability that it came from Factory A. Show how you apply Bayes’ rule.

Problem 2 (20 points in total).

Respond to each part in 2–4 sentences in your own words. When asked for an example, one well-chosen example is enough.

- (a) (5 points) Explain supervised and unsupervised learning, highlighting what information is available in each setting and what each is used for. Give one example of each.
- (b) (5 points) Explain regression and classification, highlighting their similarities and differences, including what type of response variable each involves. Give one example of each.
- (c) (5 points) Explain prediction and inference, and give one example scenario/task for each.
- (d) (5 points) Explain parametric and non-parametric methods, including their differences as well as one advantage and one disadvantage of each.

Problem 3 (25 points in total).

Use the built-in R dataset `cars`, which contains 50 observations of speed (`speed`, in miles per hour) and stopping distance (`dist`, in feet), recorded in the 1920s.¹ Model stopping distance as a function of speed:

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \varepsilon_i.$$

For the inference in (c)–(d), assume that, conditional on the observed speeds, the errors are independent $\mathcal{N}(0, \sigma^2)$. This is a working assumption for the exercise, not a conclusion established from the data.

- (a) **(6 points)** Make a scatterplot of stopping distance versus speed, with labeled axes and units. Fit the model using `lm(dist ~ speed, data = cars)`, add the fitted line to the plot, and report the fitted equation. (*Hint: plot and abline.*)
- (b) **(5 points)** Interpret the estimated slope in context, including its units. Is the estimated intercept practically meaningful, given the observed range of speeds? Report R^2 and explain what it measures for this dataset. (*Hint: summary and range.*)
- (c) **(7 points)** Test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ at the 5% significance level. Report the slope's standard error, t -statistic, p -value, and 95% confidence interval, and state your conclusion in context. Does this test by itself establish a causal effect? (*Hint: summary and confint.*)
- (d) **(7 points)** At a speed of 15 miles per hour, compute the fitted mean stopping distance, a 95% confidence interval for the mean stopping distance, and a 95% prediction interval for a single new observation from the same population. Explain the different targets of the two intervals and why one is wider. (*Hint: use predict with newdata = data.frame(speed = 15) and either interval = "confidence" or interval = "prediction".*)

Problem 4 (15 points in total + 5 bonus points).

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where the predictors x_i are fixed, $n \geq 3$, and the errors $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 > 0$.

- (a*) **(5 bonus points)** Assuming $\sum_{i=1}^n (x_i - \bar{x})^2 > 0$, prove that the minimizer of

$$\text{RSS} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

is given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

(*Hint: first minimize over β_0 for a fixed β_1 , then minimize over β_1 .*)

- (b) **(7 points)** Suppose we are given the dataset

$$\{(-3, -4), (-2, -2), (1, 0), (2, 3), (3, 5), (4, 6), (5, 8)\}.$$

¹Dataset description: R documentation for `cars`.

Use the displayed formulas in (a*) to compute $\hat{\beta}_0$ and $\hat{\beta}_1$. You may use these formulas without completing the optional proof. Show $\bar{x}$, $\bar{y}$, $S_{xx} = \sum_i (x_i - \bar{x})^2$, and $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$. You may use R or a calculator for sums and arithmetic, but show the calculation rather than reporting `lm` output alone.

- (c) (8 points) For the same dataset, compute the RSS, $\text{SE}(\hat{\beta}_1)$, and a 95% confidence interval for β_1 . State the degrees of freedom used. You may use R to obtain the t critical value. (*Hint:* $s^2 = \text{RSS}/(n-2)$ and $\text{SE}(\hat{\beta}_1) = s/\sqrt{S_{xx}}$.)

Problem 5 (25 points in total + 5 bonus points).

Consider a simple linear regression and a cubic polynomial regression:

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon.$$

Both models are fitted by ordinary least squares, with an intercept.

- (a) (5 points) Explain why the cubic model is a *multiple linear regression* model by identifying its predictors. What is linear, and what is nonlinear?
- (b) (5 points) When the models are fitted on the same training data, which training RSS is guaranteed to be no larger? Can the two RSS values be equal? Justify your answer using the relationship between the two model classes.
- (c) (10 points) Compare the models in R under the two true mean functions

$$f_1(x) = 5 + 2x, \quad f_2(x) = 0.5x^2 + 2\sin x,$$

with $X \sim \text{Uniform}(0, 10)$ and $Y = f_j(X) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, 2^2)$ is independent of X . The following starter code generates independent observations:

```
set.seed(2026)
make_data <- function(n, f) {
  x <- runif(n, 0, 10)
  data.frame(x = x, y = f(x) + rnorm(n, sd = 2))
}
f1 <- function(x) 5 + 2*x
f2 <- function(x) 0.5*x^2 + 2*sin(x)
```

For each true mean function, generate one training set of size 100 and an independent test set of size 500. Fit both models on the *same training set* and evaluate them on the *same test set*; do not fit models to the test data. Report one four-row table with columns: true mean function, fitted model, training MSE, and test MSE, where

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

for an evaluation set of size m . No simulation loop or plots are required.

(*Hints:* use `y ~ x + I(x^2) + I(x^3)` for the cubic model and `predict` for test predictions.)

- (d) (5 points) Interpret the table in 3–5 sentences, distinguishing training fit from predictive performance. For each true mean function, identify which model has smaller training MSE and which has smaller test MSE. Does the model with smaller training MSE also have smaller test MSE in your results? Explain why these results do not establish that either model is always better.
- (e*) (5 bonus points) For f_1 only, repeat part (c) 50 times with fresh independent training and test sets. Report the average training and test MSE for each model and briefly relate the results to part (d).